

Total No. of Questions : 5]

SEAT No. :

PD-2846

[Total No. of Pages : 2

[6432]-2001
M.C.A. (Management)
JPR 551 MJ: Java Programming
(2024 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 50

Instructions to the candidates:

- 1) All questions are compulsory.*
- 2) Figures to the right indicate full marks.*

Q1) Solve any two :

- a) Define a class calculation to implement method overloading for addition of two integers and two double variables. [5]
- b) Define class person with suitable data member and methods. And extend this class in Manager class. Display manager details. [5]
- c) Write a function using lambda expression to calculate power of number.(x^y) [5]
- b) What is garbage collection? Explain with required method. [5]

Q2) Solve any two :

- a) Create a thread to display prime numbers between 1 to 500 each number will display after 3 seconds. [5]
- b) Write a java program to demonstrate how to create user-defined exception. [5]
- c) Write a Java program to implement arraylist with function to add, remove & sort member. [5]
- b) Differentiate between checked and unchecked exception. [5]

P.T.O.

Q3) Solve any one :

- a) Create a HTML page to accept two numbers and write servlet to add given numbers and display result. **[10]**
- b) Explain servlet life cycle and demonstrate its methods with example. **[10]**

Q4) Solve any one :

- a) Design Java Application (using JSP) for registration of hackthon (student name, Mail Id, Phone No, Gender, Course) and store data in appropriate table. **[10]**
- b) Explain JSP directives with example. **[10]**

Q5) Solve any one :

- a) Create a spring MVC form to read registration details for Blood donation camp with spring validation & display it. **[10]**
- b) Explain spring MVC architecture. And also explain any two spring annotations. **[10]**

